

Current conclusion display preservation

Cyclic connection and flatness

$$\Omega_u = -C_\chi(t)/u^2 + B_\chi/u, \quad \Pi_u = \Pi\Omega_u, \quad \Omega_t = -A(t)/u, \quad \Pi_t = \Pi\Omega_t, \quad \partial_u\Omega_t - \partial_t\Omega_u + [\Omega_u, \Omega_t] = 0.$$

Residual collision annihilator and exact modules

$$\psi_{h,k}(X) = \prod_{\mu} (X - \mu)^{\max_{\alpha}(\sum_i \rho_i) = \mu \min(p, 1 + \sum_i (m_i - 1))}, \quad \ker \bar{\eta}_k = (\psi_{h,k})/(\bar{\chi}_{h,k}), \quad \text{im } \bar{\eta}_k \simeq k_0[X]/(\psi_{h,k}).$$

Translated metric and all three pure derivatives

$$\psi_a - \psi = 2p \operatorname{Re}(h_a(t)/u), \quad \partial_u(\psi_a - \psi) = -ph_a(t)/u^2, \quad \partial_u^2(\psi_a - \psi) = 2ph_a(t)/u^3, \quad \partial_u \partial_t(\psi_a - \psi) = pa/u^2.$$

Existing parent RMT16 overlay (unchanged)

$$\begin{aligned} \mathcal{H}_a(B) &= 2 \operatorname{arcosh}(e^{B/(2a)}), & \mathcal{H}'_a(B) &= \frac{1}{a\sqrt{1 - e^{-B/a}}}, \\ \mathcal{H}_a^{-1}(z) &= 2a \log \cosh(z/2) \quad (z > 0). \end{aligned} \tag{RMT16}$$